FLOCK ENERGY / DATA SCIENCE TAKE-HOME

Washing-Machine Load Disaggregation from REFIT Data

A reproducible data-cleaning, usage-analysis and machine-learning proof of concept

Problem	Estimate washing-machine power from a whole-home smart-meter signal.
Data	Official REFIT raw House 1 + cleaned Houses 1, 2, 3, 5.
Method	Minute-level quality audit, cycle EDA, always-off baseline, Random Forest.
Validation	Disjoint household split: train 1/2, validate 3, test 5.
Evidence status	REFIT RESULTS INCLUDED

Important: The assignment requires a **public GitHub repository** with executable code and four specific Markdown deliverables. This PDF is the companion report. A PDF alone does not meet the stated submission instructions.

Evidence: Plots and statistics in this report were generated from a run labelled official REFIT; verify source file provenance and visually inspect plots before publishing.

Prepared with Python, Pandas, scikit-learn, Matplotlib and ReportLab | 2026

01 The assignment explained in plain English

A household electricity meter measures the sum of all appliances operating at that moment. A utility usually does not know whether a washing machine, kettle or dishwasher caused an increase. The challenge is to infer washing-machine power from the aggregate signal; this is known as non-intrusive load monitoring (NILM).

Illustrative example (NOT a measured REFIT result)	Power
Refrigerator	150 W
Washing machine	1,800 W
TV, fan, and lights combined	250 W
Total observed by whole-house meter	2,200 W

The model sees the 2,200 W whole-house input. During training it also has a correctly labelled washing-machine submeter channel; during inference it must predict the washing-machine signal without that appliance channel.

What the four assignment sections ask for

Required stage	Deliverable in this implementation
1. Raw cleaning	Audit timestamps, duplicates, nonphysical data, missing minutes; clean raw House 1; make before-and-after plots.
2. Usage exploration	Four homes; washer sensor mapping; candidate cycles; hourly/weekly patterns; cross-home figures.
3. Model and evaluation	Aggregate-only baseline and RF model; split by household; active/off metrics and actual-versus-predicted plot.
4. Utility insight	Customer-facing off-peak opportunity, success criteria, transfer limitations, coarser meter data.

What matters most: correct data handling and reasoning. The brief explicitly prefers a simple correct analysis to an advanced model that is difficult to justify.

02 Official dataset and study design

REFIT is a University of Strathclyde electricity dataset from 20 UK households, collected across an overall 2013-2015 period. Original readings are about every 8 seconds and are measured in watts. Each home includes an aggregate channel and up to nine appliance submeter channels. Exact availability differs by home; individual study dates are determined from executed data.

Two dataset editions are deliberately used: the original raw edition for the House 1 quality audit, and the official cleaned edition for cross-home washing-machine analysis and modelling. The latter has already undergone special corrections, including daylight-saving adjustments and imputation.

House	Official washer channel	Purpose
1	Appliance5	Raw + cleaned; training
2	Appliance2	Cleaned; training
3	Appliance6	Cleaned; validation
5	Appliance3	Cleaned; held-out test

This mapping is not optional. A fixed Appliance5 target across all homes would train on the wrong appliances for Houses 2, 3 and 5. House 4 has two monitored washers and is excluded to keep this example unambiguous.

Analysis window and units

- Default requested study period: 2014-09-01 to 2014-12-01, first date inclusive and second exclusive, in UTC.
- Resample individual readings to 1-minute mean power, not the sum of watts.
- Energy of a 1-minute sample: kWh = power in W / 60,000. Sum this over minutes of a detected cycle.
- Official cleaned files may contain artificial zeros used for long outages; a zero can therefore mean missingness instead of genuine appliance inactivity.

03 Raw data cleaning: exact decisions

Problem	Professional response
Unparseable timestamps	Parse Unix epoch in UTC where present. Count and exclude timestamps that cannot be parsed.
Duplicate timestamps	Stable rule: last record in source file wins, then sort ascending.
Irregular measurements	Compute differences between consecutive timestamps; log intervals exceeding 20 seconds.
Negative values / high IAM spikes	Invalidate negative power; invalidate appliance readings above 4000 W. Note aggregate readings over 25 kW for manual review.
Small missing gaps	Only fill entire missing runs of one or two minutes if bounded by valid observations.
Long outages	Preserve NaN throughout the outage, not zero or partially filled values.
Official Issues flag	If available, discard minutes with a submeter-sum inconsistency.

Why not fill every missing gap?

A 90-minute sensor outage does not mean the washing machine was turned off for 90 minutes. Replacing missing values with zero creates false evidence, biases energy totals and may teach the model that a genuine power-off event occurred.

Why the one-minute mean?

The assignment requests 1-minute power estimates. Average instantaneous power samples within each minute to produce one point per minute. Only afterward convert watts to kWh using the duration. Summing raw watts directly is not a meaningful energy calculation.

04 Exploratory analysis: detect washer cycles

An operating event is not every individual spike: washing machines often pause, fill, agitate and spin. The code uses a transparent heuristic rather than pretending to have labelled ground-truth cycle boundaries.

Rule parameter	Chosen working value
Appliance considered active	Measured power >20 W
Permitted internal pause	Up to 10 minutes, if all intervening minutes are observed
Minimum total event span	15 minutes
Minimum above-threshold samples	3 one-minute points

These are initial engineering assumptions, **not** manufacturer-verified cycle labels. A robust future analysis should compare cycle counts across thresholds (e.g. 10, 20 and 50 W) and inspect visually identified cycles.

Questions answered for each home

• How many valid/missing minutes were recorded? What percentage of observed washing-machine minutes were off?

• What are the actual coverage start and end dates, and how many candidate cycles were found?

• At which hours and weekdays do events begin? The plotting code uses UTC, so local UK behavioural interpretation needs time-zone conversion.

• Which homes have larger median cycle duration and estimated energy per detected event?

The six output figures include raw versus cleaned signals, a 24-hour whole-home/washing-machine trace, one detected cycle, cross-home comparison, usage timing, and held-out test actual-versus-predicted power.

05 Model and honest evaluation

Target: washing-machine power W at minute t. Features: whole-home aggregate power W at t and earlier minutes, its trailing rolling mean/standard deviation, lagged changes, and calendar encodings. The model never receives the washer submeter as an input feature.

Model	Why included
Always-off baseline (0 W)	Necessary because off minutes dominate, and low average error may be misleading.
Random Forest regression	Works with nonlinear power patterns and numeric engineered features; straightforward to implement and explain.

Split before inference: keep whole households apart

Training = House 1 + House 2. **Validation** = House 3. **Untouched final test** = House 5. This tests generalisation to a household that was not part of model fitting. Randomly splitting adjacent one-minute readings would leak very similar, overlapping cycles between train and test.

Evaluation metrics and their meanings

Metric	Interpretation
MAE (W)	Mean magnitude of watt-level error; compare with always-off baseline.
RMSE (W)	Larger errors are penalised more heavily; detects missed high peaks.
Active-minute MAE (W)	Error when washing machine truly operated; avoids hiding errors in off time.
Active precision / recall / F1	Can the predicted watts correctly mark operating minutes (>20 W)?
Energy bias (%)	Compare predicted with actual accumulated kWh for observed minutes.

A Random Forest is not automatically better than the always-off baseline. If it loses, report it and discuss why. Do not tune to House 5 test results; any changes should use training/validation only.

06 Realistic customer and utility recommendation

Proposed insight

Use inferred laundry timing and approximate energy to give interested customers optional information: "Your washing-machine use is estimated to occur mostly around these hours. If you have a suitable time-of-use tariff, running it later could reduce peak-hour costs." Confirm tariff eligibility and display uncertainty.

How to prove value

- Verify accuracy against real appliance submeter readings in a ground-truth sample.
- Run a randomised opt-in pilot, with a comparison group receiving no personalised washing-machine advice.
- Track peak-period kWh shifted, energy and bill changes, opt-in/opt-out rates, customer satisfaction and erroneous event notifications.
- Make a distinction: shifting a wash to off-peak does not automatically reduce the kWh used.

Where the system can fail

- Two devices switch on at once; their electrical signatures overlap.
- A new household owns a different washer with a previously unseen heating/spinning pattern.
- Official prefilled zeros, communication dropouts or inconsistent sensor channels make some labels unreliable.
- UK data should not be deployed to Indian customers without India-specific sampling, tariff, appliance and validation evidence.

What changes with 15-minute or 30-minute meter data?

A five-minute high-power phase is averaged with many low-power minutes. Peaks, event boundaries and short pauses disappear, so reconstructing washing-machine cycles accurately becomes substantially harder. Summary-level behavioural signals might remain, but measured 1-minute model performance cannot be assumed to transfer to 15/30-minute data.

07 Reproduction and exact submission instructions

Step A - install and test

Use Python 3.10+. From the extracted project directory, create a virtual environment, run **pip install -r requirements.txt**, then **pip install -e .**, and finally **pytest -q**.

Step B - download and extract REFIT

Download REFIT_RAW_081116.7z and CLEAN_REFIT_081116.7z from the official URLs in README.md. Extract raw CSVs under data/raw and official cleaned CSVs under data/cleaned. The raw and cleaned compressed archives are several hundred megabytes each; do not upload the dataset into GitHub.

Step C - run the project on measured data

```
python -m refit_nilm.pipeline --raw-dir data/raw --clean-dir data/cleaned
--output-dir outputs --start 2014-09-01 --end 2014-12-01
```

Step D - inspect and finalise

Open every PNG in outputs/figures. Check coverage, quality flags, event reasonableness and the baseline/model metrics. Then run **python scripts/finalize_submission.py --output-dir outputs** to populate the required data-backed Markdown files and generate a submission PDF. It intentionally refuses synthetic demo runs.

Step E - submit the GitHub repository

Required file	What it contains
Python source code	Executable ingestion, cleaning, EDA, prediction, evaluation and tests
README.md	Reproduction, assumptions, features, data links and AI assistance disclosure
DATA.md	Real raw-quality observations and cleaning decisions
RESULTS.md	Real EDA, results, validation, failures and plotted evidence
Recommendation.md	One-page utility summary and section 4 answers
Optional PDF	Full report, with genuine REFIT results only after actual execution

08 Measured REFIT experiment outputs

Raw House 1 selected-period quality: **1,400,757 input records**; **170,622 duplicate timestamps**; **0 invalid timestamps**; **3,334 unobserved minute bins**.

House	Channel	Observed minutes	Missing minutes	Candidate cycles	Median kWh/event
1	Appliance5	124,390	5,635	17	0.0399
2	Appliance2	120,576	10,464	99	0.0704
3	Appliance6	121,682	8,343	77	0.0653
5	Appliance3	117,080	13,960	131	0.0801

Model performance, evaluated without test-house training

Split	Method	MAE (W)	RMSE (W)	Active F1	Energy bias %
validation H3	always_off	4.3723	76.5287	0.0	-100.0
validation H3	random_forest	14.2884	96.1694	0.2051	180.9
test H5	always_off	10.4598	79.0333	0.0	-100.0
test H5	random_forest	18.2012	83.3441	0.1477	7.38

Metric interpretation: overall MAE can favour always-off due to infrequent operation; review active MAE, active F1 and individual cycles before concluding the model is useful.

09 REFIT plots: raw/cleaned and usage



Figure 1. Actual raw versus processed House 1 signals (same UTC day), for aggregate and two appliance channels.

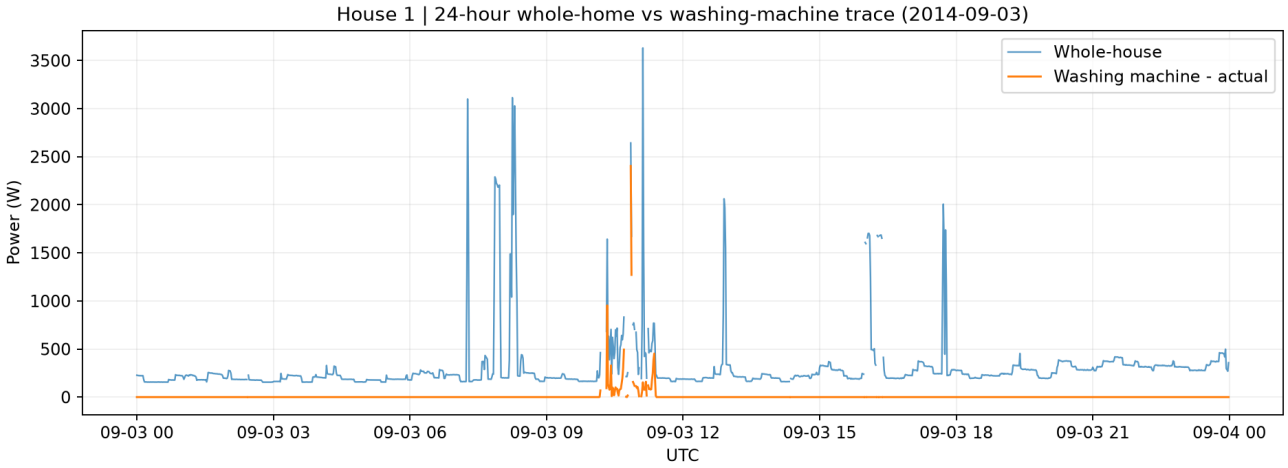


Figure 2. One 24-hour reference trace, measured aggregate and washer.

10 REFIT plots: cycles and cross-home behaviour

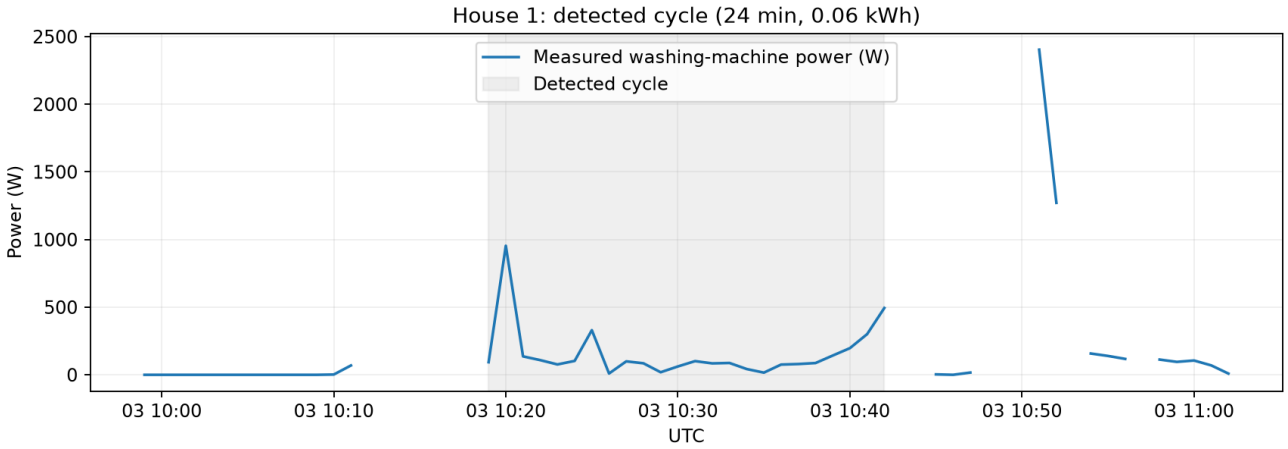


Figure 3. Example inferred washing-machine event; event boundaries are heuristic.

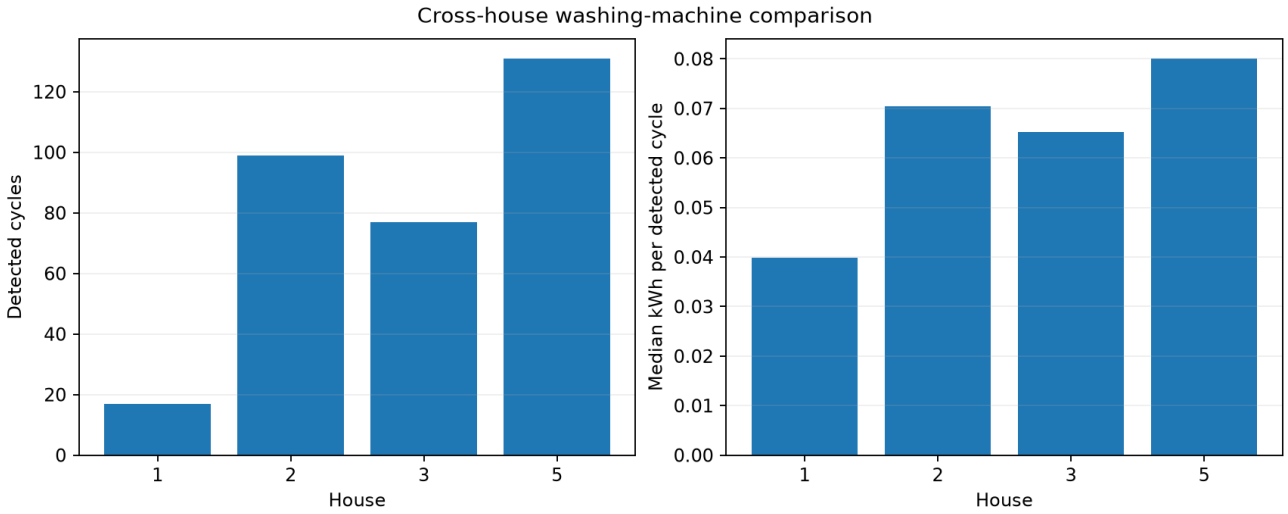


Figure 4. Cross-house event count and median energy per candidate event.

11 REFIT plots: usage patterns and test prediction

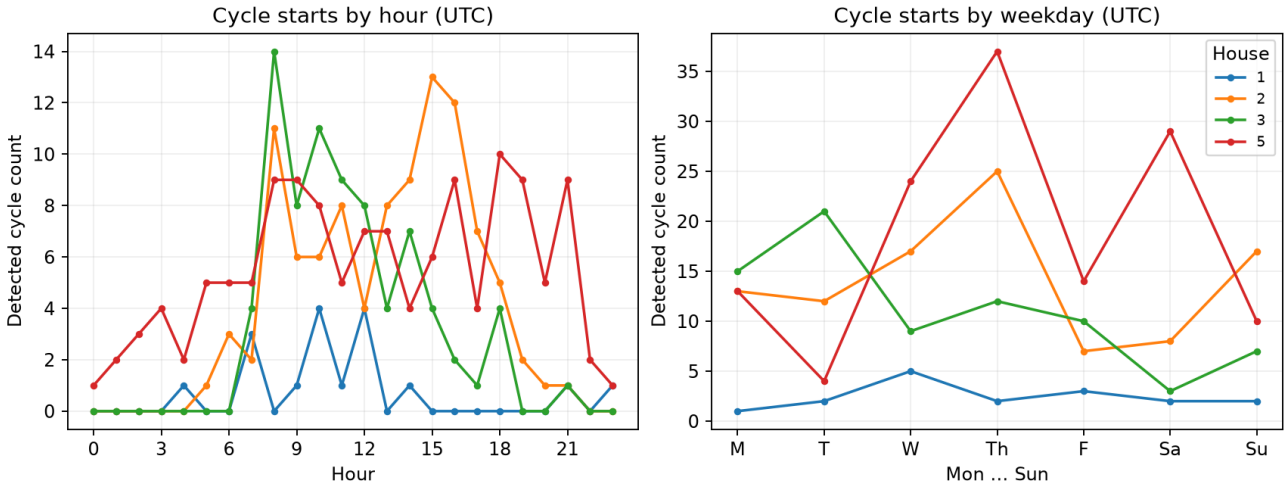


Figure 5. Cycle starts by UTC hour and day of week across four houses.

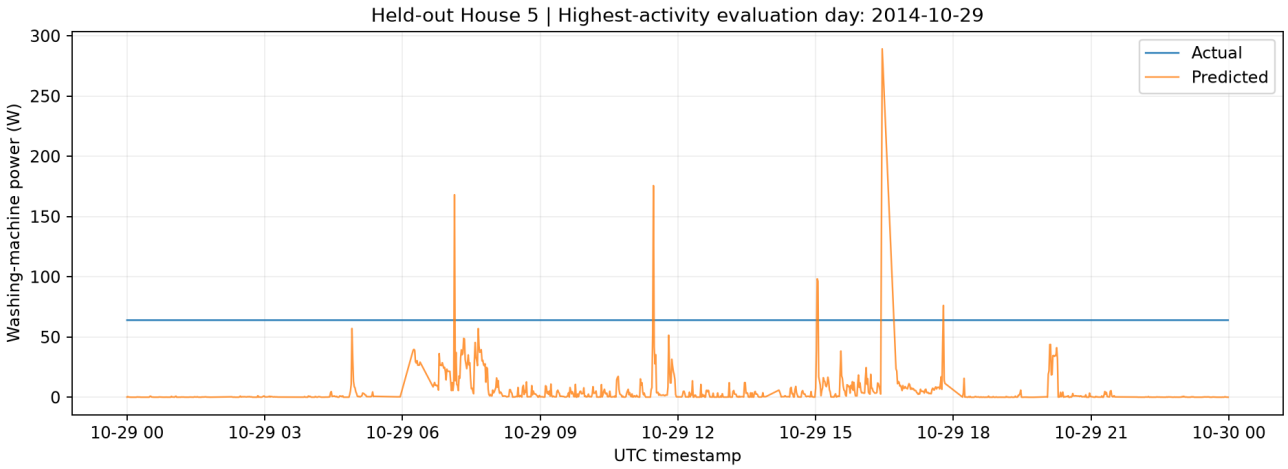


Figure 6. Actual washer versus RF prediction on the untouched House 5 test set.

References, attribution and responsible use

• University of Strathclyde, REFIT: Electrical Load Measurements (Raw). https://pureportal.strath.ac.uk/en/datasets/refit-electrical-load-measurements/

• University of Strathclyde, REFIT: Electrical Load Measurements (Cleaned). https://pureportal.strath.ac.uk/en/datasets/refit-electrical-load-measurements-cleaned/

• Official cleaned README: https://pure.strath.ac.uk/ws/portalfiles/portal/62090183/CLEAN_READ_ME_081116.txt

• Murray, D., Stankovic, L., and Stankovic, V. (2017). An electrical load measurements dataset of United Kingdom households from a two-year longitudinal study. Scientific Data, 4:160122. https://doi.org/10.1038/sdata.2016.122

• Dataset licence: Creative Commons Attribution 4.0 (CC BY 4.0).

What the candidate should say about AI assistance

ChatGPT helped scaffold the Python implementation, draft the explanatory documentation, write unit tests and structure this PDF. The candidate must execute the measured-data experiment, verify every plot, interpret results, document edits, and be prepared to explain all reasoning. Do not present an AI-generated description as personal experimental observation.

Suggested technical progression

This is fundamentally a Data Science / Machine Learning Engineering assignment. The high-value professional skill is the combination of careful data engineering, quantitative evaluation, and reproducible code. A web frontend or elaborate deployment system is not requested and would add risk without strengthening this submission.

Appendix A | Post-baseline experiments and fresh holdout

Scope: This appendix updates the main report. The original House 5 plots and test numbers document the initial benchmark. After those results were inspected, additional experiments were selected on House 3 validation and evaluated once on House 6. House 5 is not described as untouched for the later experiments.

A1. Model selection on House 3

Ablations investigated removing calendar features, a 50 W regression output cutoff, and a two-stage activity classifier plus active-only power regressor. The predeclared selection criterion for the two-stage comparison was maximum **minute-level F1** on House 3, with MAE as tie-break. The selected method was **Original RF + 50W cutoff**.

House 3 candidate	MAE (W)	Active F1	Recall	Bias (%)
Always off	4.37	0.000	0.000	-100.00
Original RF + 50W cutoff	11.77	0.216	0.427	115.85
Original RF + 50W cutoff	11.77	0.216	0.427	115.85

The two-stage approach was rejected: its explored probability cutoffs all had poorer validation F1 than the selected original Random Forest + 50 W cutoff. No evidence supports claiming that the two-stage classifier improved the problem.

A2. House 6 independent evaluation

The selected configuration was locked before House 6 evaluation. House 6 uses the official cleaned REFIT **Appliance2** washing-machine channel. It has **93,695** model-ready one-minute readings over the same date range.

House 6 method	MAE (W)	Active MAE	Precision	Recall	F1	Bias (%)
Always off	3.12	319.20	0.000	0.000	0.000	-100.00
VALIDATION-SELECTED: Original RF + 50W cutoff	5.39	237.75	0.217	0.500	0.303	+27.45

The model detected half the active minutes (recall 0.500), but precision was only 0.217. Active-minute MAE improved from 319.20 W to 237.75 W; however overall MAE **worsened** from 3.12 W to 5.39 W. Estimated total energy was +27.45% relative to observed energy. These findings do not justify customer-facing appliance-specific claims.

A3. Limitations and integrity

House 3 was consulted repeatedly during the investigation; its validation score may be optimistic. House 5 motivated subsequent analysis and is not a fresh test for those later changes. House 6 was evaluated after freezing the selected method; do not repeatedly tune on House 6. Results apply only to the selected three-month REFIT subset, 1-minute averages, and the documented washing-machine labelling assumptions.

The main report documents raw-data quality, cycle detection and household comparison. The appendix documents post-baseline evaluation, including negative results. Preserve the original results.json and original trained model for auditability.

Appendix B | House 6 actual versus predicted trace

Illustrative day selected from the House 6 holdout to show measured and inferred washing-machine power. This figure is not a separate performance test; numeric metrics above cover the full holdout.

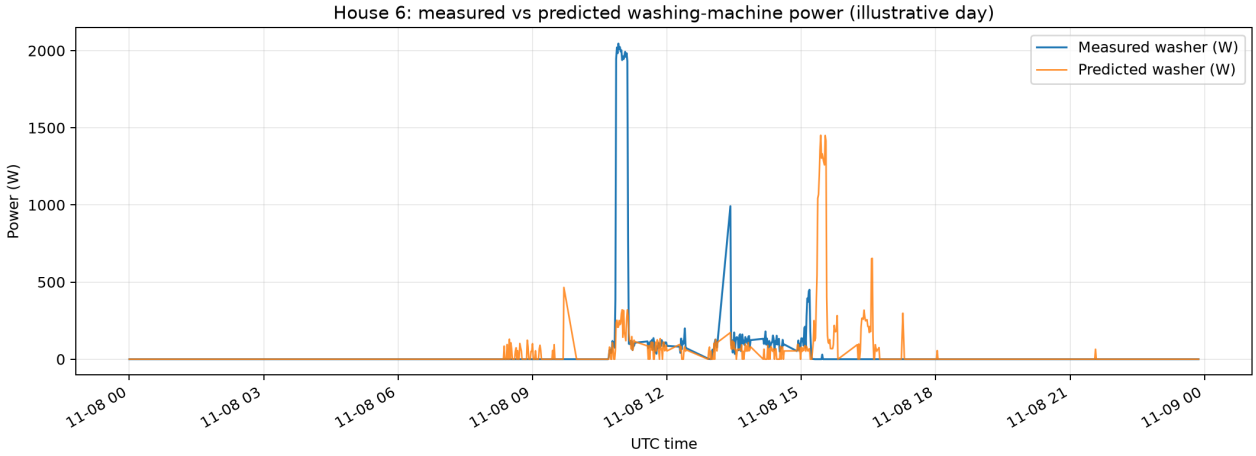


Figure A1. House 6 observed washing-machine watts and selected-model prediction.

Interpretation: Assess both missed peaks and false positive bursts; aggregate-level resemblance alone is insufficient evidence of accurate appliance detection.